

# Dr. Fabrizio Gentile

✉ fabrizio.gentile@cea.fr

ORCID: 0000-0002-8008-9871

## Work Experience

- 2024 - today  **Postdoctoral researcher**  
CEA Paris - Saclay  
**Advisors:** David Elbaz (CEA) & Emanuele Daddi (CEA)
- 2021 - today  **Associate astronomer**  
INAF - Osservatorio di Astrofisica e Scienza dello Spazio (Bologna)
- 2023 - 2024  **Visiting scholar (Marco Polo Scholarship funded by UniBo)**  
The University of Texas at Austin  
**Project title:** *Unveiling the dark Universe with JWST*  
**Supervisor:** Caitlin Casey (UT Austin)

## Education

- 2021 - 2025  **Ph.D. in Astrophysics [*cum laude*]**  
Alma Mater Studiorum - University of Bologna  
**Thesis title:** *Searching for high-redshift progenitors of massive galaxies*  
**Supervisors:** Margherita Talia (UniBo) & Andrea Cimatti (UniBo)
- 2019 - 2021  **M.Sc. in Physics (Astrophysics Curriculum) [110/110 *cum laude*]**  
University of Naples "Federico II"  
**Thesis title:** *Fast Automated Analysis of strong gravitational lenses in the forthcoming Euclid survey using machine-learning methods*  
**Supervisors:** Crescenzo Tortora (INAF) & Giovanni Covone (UniNa)
- 2016 - 2019  **B.Sc. in Physics [110/110 *cum laude*]**  
University of Naples "Federico II"  
**Thesis title:** *Searching for strong gravitational lenses with Convolutional Neural Networks*  
**Supervisors:** Crescenzo Tortora (INAF) & Giovanni Covone (UniNa)

## Research

- Scientific interests*  Galaxy formation and evolution • IR and (sub)mm astronomy • High-z galaxies  
• Strong gravitational lensing • Multi-wavelength surveys • Astrophysical applications of machine learning
- International collaborations*  Member of the **COSMOS** and **COSMOS-Web** collaborations  
 Member of the **Euclid Consortium** (Galaxy Evolution and Strong gravitational lensing Science Working Groups)  
 Member of the **ELSA** (Euclid Legacy Science Advanced Analysis Tools) project  
 DPo Delegate for the **Vera Rubin Observatory** collaboration

## Research (continued)

- Invited talks*
- *"Seeing the unseen: a radio selection for high- $z$  dusty galaxies" (The growth of galaxies in the early Universe, Sexten-Sesto (IT), January 2026)*
  - *"Galaxy quenching and morphological transformation: different evolutionary paths in the field and in high-density environments with Euclid Q<sub>1</sub> data" (Working with Euclid Q<sub>1</sub>, Madrid (ES), December 2025)*
  - *"Massive and dusty galaxies: a radio + JWST perspective" (Big Galaxies, Big Problems, Leiden (NL), April 2025)*
- Contributed talks*
- *"Galaxy quenching and morphological transformation: different evolutionary paths in the field and in high-density environments with Euclid Q<sub>1</sub> data" (ELSA Collaboration Meeting, Bologna (IT), February 2026)*
  - *"Galaxy quenching and morphological transformation: different evolutionary paths in the field and in high-density environments with Euclid Q<sub>1</sub> data" (Euclid France 2025, Strasbourg (FR), November 2025)*
  - *"ALMA Vs JWST: What's the best way of selecting dusty galaxies?" (COSMOS Meeting 2025, Marseille (FR), May 2025)*
  - *"How to form and destroy your (massive) disk" (Euclid from Q<sub>1</sub> to DR<sub>1</sub>, Tenerife (ES), January 2025)*
  - *"Form and substance in galaxy quenching: A first analysis with Euclid Q<sub>1</sub> Data" (ELSA Collaboration Meeting, Bristol (UK), January 2025)*
  - *"Dark progenitors and massive descendants: An ALMA/JWST perspective on Radio-Selected NIRdark galaxies" (ALMABO 2024, Bologna (IT), September 2024)*
  - *"Radio-Selected NIRdark galaxies in COSMOS-Web: A JWST view behind the dust" (COSMOS Meeting 2024, Kashiwa (JP), July 2024)*
  - *"Dark progenitors and massive descendants: an ALMA/JWST perspective on Radio-Selected NIRdark galaxies" (Star Formation 2024, Cambridge (UK), April 2024)*
  - *"Radio-Selected NIRdark galaxies: likely progenitors of high- $z$  massive galaxies?" (EAS2023, Cracow (PL), July 2023)*
  - *"Radio-Selected NIRdark galaxies: the ALMA view behind the dust" (Observing the mm Universe, Grenoble (FR), June 2023)*
  - *"An ALMA/JWST look into Radio-Selected NIRdark galaxies" (COSMOS Meeting 2023, Rochester (US), May 2023)*
  - *"Radio-Selected NIRdark galaxies: the ALMA view behind the dust" (ALMA Meeting for Young Astronomers, Online, March 2023)*
  - *"Bayesian Neural Networks and strong lensing: towards an uncertainties-aware machine learning algorithm for lens-modelling" (EAS2022, Valencia (ES), June 2022)*
- Invited seminars*
- *"Galaxy quenching and morphological transformation: different evolutionary paths in the field and in high-density environments with Euclid Q<sub>1</sub> data" (DAWN Center, Copenhagen (DK), December 2025)*
  - *"Seeing the Unseen: A multi-wavelength quest for optically-dark dusty galaxies" (CEA Paris-Saclay, Saclay (FR), May 2025)*
  - *"Not-so-little red dots: red and massive sources in the first COSMOS-Web data" (National Observatory of Athens, Online, January 2025)*
  - *"Radio-Selected NIRdark galaxies: likely progenitors of high- $z$  massive galaxies?" (INAF-OACN, April 2023)*

## Research (continued)

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | ■ <i>"Strong gravitational lensing: a machine learning perspective" (INAF-OACN, April 2022)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Competitive telescope time | <div>■ <b>PI:</b> <i>"Illuminating the dark side of star formation at <math>z &gt; 3</math>" (ALMA, 27.4h)</i></div> <div>■ <b>PI:</b> <i>Disky titans in high-density environments challenging quenching models" (NOEMA, 20.0 h)</i></div> <div>■ <b>Co-I:</b> <i>"Brighter than GN-z11? Grism Observations of the brightest <math>z \sim 11</math> Candidates found in COSMOS-Web" (HST, 24 orbits)</i></div> <div>■ <b>Co-I:</b> <i>"ALMA Data for Every Sub-Millimeter Galaxy Over the COSMOS-Web and COSMOS-3D Survey Fields" (ALMA, 4.9h)</i></div> <div>■ <b>Co-I:</b> <i>"ALMA follow-up of Little Red Dots: Efficiently testing black hole growth/feedback models with multiphase gas kinematics" (ALMA, 12.8h)</i></div> <div>■ <b>Co-I:</b> <i>"A comprehensive population study of Little Red Dots: Connecting early BH and galaxy growth" (JWST, 86.6 h)</i></div> <div>■ <b>Co-I:</b> <i>"The obscured Universe at <math>z \sim 6</math> on the way to JWST" (NOEMA, 27h)</i></div> <div>■ <b>Co-I:</b> <i>"The COSMOS High-<math>z</math> ALMA-MIRI Population Survey (CHAMPS): A Wide-Area Comprehensive Survey of the Dusty Universe" (ALMA, 143.5h)</i></div> <div>■ <b>Co-I:</b> <i>"Caught in the Web: ALMA Data for Every Sub-Millimeter Galaxy Over the COSMOS-Web Survey Field" (ALMA, 6.7h)</i></div> |
| Competitive grants         | <div>■ <b>Co-I:</b> <i>"A JWST Archival Analysis of the Most Dust-Obscured Galaxies Across Cosmic Time" (JWST AR Program, 2024)</i></div> <div>■ <b>Co-I:</b> <i>"Mass profiles of gravitational Lenses: MOdelling with Neural networks and scientific exploitation" (INAF Mini-Grant, 2022)</i></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Visiting periods           | ■ <b>Austin, TX, USA (Oct 2023 - Apr 2024):</b> Visiting the University of Texas at Austin to work with Dr. Caitlin Casey on the project "Unveiling the dark universe with JWST".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Other relevant activities  | <div>■ <b>Co-Supervisor</b> of Matteo Saponi's MSc thesis at University of Bologna</div> <div>■ <b>Referee</b> for <i>The Astrophysical Journal Letters</i>, <i>The Astrophysical Journal</i>, <i>Astronomy &amp; Astrophysics</i>, <i>Monthly Notices of the Royal Astronomical Society</i>, and <i>Open Research Europe</i>.</div> <div>■ <b>Member of the LOC</b> for the conference "Views on the multi-phase interstellar medium in galaxies" (Bologna, September 2024)</div> <div>■ <b>Organizer</b> of the group meetings of the LCEG laboratory at CEA Paris-Saclay</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Teaching and outreach

|                     |                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching experience | ■ <b>Exercitation of mechanics &amp; thermodynamics</b> ( <i>Bachelor level</i> )<br>University of Naples "Federico II"<br>150 students - 50h (A.Y. 2021-22) |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Teaching and outreach (continued)

|                     |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outreach experience | ■ <b>Ideation and performing of astronomy experiments for school students</b><br>Astronomy Festival " <i>The Universe in all senses</i> "<br>Castellaro Lagusello (Italy), 2023                                                                                                                                                                                    |
| Press releases      | ■ <b>CEA Paris-Saclay:</b> Two press releases on my paper with <i>Euclid</i> Q1 data ( <a href="#">Link 1</a> and <a href="#">Link 2</a> )                                                                                                                                                                                                                         |
| and media presence  | ■ <b>Euclid Consortium:</b> Press release on my paper with <i>Euclid</i> Q1 data ( <a href="#">Link</a> )<br>■ <b>American Astronomical Society:</b> Video-interview for the <i>AAS Journal Author Series</i> on the "Not-so-little Red Dots"( <a href="#">Link</a> )<br>■ <b>Media INAF:</b> Press-release for a paper on the LEMON code ( <a href="#">Link</a> ) |

## Skills

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages        | ■ Italian (native) • English (proficient) • French (intermediate)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Computer skills  | ■ <b>Main developer</b> of two public codes for astrophysical analyses: LENS MOdelling with Neural networks (LEMON) and Photometry Extractor for Blended Objects (PhoEBO)<br>■ Extensive use of the Windows, Linux/Unix, and MacOS Operative Systems<br>■ Advanced knowledge of Python 3 and its main libraries concerning astronomy (astropy, photutils), deep learning (keras, tensorflow), plotting (matplotlib, seaborn) and data analysis (numpy, scipy, pandas, emcee)<br>■ Good knowledge of C, C++ and SQL<br>■ Extensive use of the Microsoft Office suite<br>■ Extensive use of L <sup>A</sup> T <sub>E</sub> X and Overleaf |
| Technical skills | ■ Photometry extraction with sEXTRACTOR, PHOTUTILS and THE FARMER<br>■ Analysis of ALMA and NOEMA interferometric data with CASA (proficient) and GILDAS (intermediate)<br>■ SED-Fitting with MAGPHYS, CIGALE, EAZY, BAGPIPES, and DENSE BASIS<br>■ Implementation of machine learning algorithms with KERAS and TENSORFLOW<br>■ Experience with SWARP, PSFEX, and analogous softwares                                                                                                                                                                                                                                                 |

## Publications

- Total number of **peer-reviewed publications**: 38 (7 as first author). Total number of **citations** in ADS: 2.000 (109 for first-author papers). **H-index**: 17. ([ADS Summary](#))

### Submitted

- **Billand et al., (2026b)**: "Do little red dots really form a distinct class of astronomical objects?" (**Submitted to A&A** - [ADS Link](#))
- **D'Eugenio et al., (2026)**: "A first [CII] view of high-z quiescent galaxies" (**Submitted to A&A** - [ADS Link](#))
- **Elias et al., (2026)**: "Quiescent fractions in high-redshift galaxy groups reflect their hot-or-cold state of gas accretion" (**Submitted to A&A** - [ADS Link](#))
- **Euclid Collaboration: Dusserre et al. (2026)**: "Euclid Quick Data Release (Q1). The first Euclid view of galaxy protocluster candidates at cosmic noon" (**Submitted to A&A** - [ADS Link](#))
- **Euclid Collaboration: Giulietti et al. (2026)**: "Euclid Quick Data Release (Q1). Searching for radio-selected Euclid-dark galaxies in the EDF-N" (**Submitted to A&A** - [ADS Link](#))
- **Franco et al. (2025b)**: "Physical properties of galaxies and the UV Luminosity Function from  $z \sim 6$  to  $z \sim 14$  in COSMOS-Web" (**Submitted to ApJ** - [ADS Link](#))

### Accepted

- **Euclid Collaboration: Gentile et al. (2025)**: "Euclid Quick Data Release (Q1). Quenching precedes bulge formation in dense environments but follows it in the field" (**Accepted by A&A** - [ADS Link](#))
- **Euclid Collaboration: Busillo et al. (2025)**: "Euclid Quick Data Release (Q1). LEMON – Lens Modelling with Neural networks. Automated and fast modelling of Euclid gravitational lenses with a singular isothermal ellipsoid mass profile" (**Accepted by A&A** - [ADS Link](#))
- **Euclid Collaboration: Enia et al. (2025)**: "Euclid Quick Data Release (Q1). A first view of the star-forming main sequence in the Euclid Deep Fields" (**Accepted by A&A** - [ADS Link](#))
- **Mancini et al.**: "A SHARP Look at Quenching and Bulge–Disk Growth in Massive Galaxies at Cosmic Noon" (**Accepted by New Astronomy**)
- **Mazzolari et al., (2026)**: "Radio selection of heavily obscured AGN in the J1030 field: unraveling a missing Compton-thick population" (**Accepted by A&A** - [ADS Link](#))

### 2026

- **Billand et al., (2026a)**: "Investigating the Growth of Little Red Dot Descendants at  $z < 4$  with the JWST" In *Astronomy & Astrophysics, Volume 706, id.A29, 15 pp.* ([ADS Link](#))
- **Franco et al. (2026)**: "COSMOS-Web: Comprehensive Data Reduction for Wide-Area JWST NIR-Cam Imaging" In *The Astrophysical Journal, Volume 999, Issue 2, id.200, 18 pp.* ([ADS Link](#))
- **Ghaffari et al. (2026)**: "Star formation quenching precedes morphological transformation in COSMOS-WEB's richest galaxy groups" In *Astronomy & Astrophysics, Volume 708, id.A129, 18 pp.* ([ADS Link](#))
- **Long et al. (2026)**: "The Extended Mapping Obscuration to Reionization with ALMA (Ex-MORA) Survey:  $5\sigma$  Source Catalog and Redshift Distribution" In *The Astrophysical Journal, Volume 999, Issue 1, id.47, 20 pp.* ([ADS Link](#))
- **Shuntov et al. (2025b)**: "The stellar mass function of quiescent and star-forming galaxies and its dependence on morphology in COSMOS-Web" In *Astronomy & Astrophysics, Volume 707, id.A391, 19 pp.* ([ADS Link](#))

## Publications (continued)

- **Toni et al. (2026):** "COSMOS-Web galaxy groups: Evolution of red sequence and quiescent galaxy fraction" In *Astronomy & Astrophysics*, Volume 707, id.A87, 19 pp. ([ADS Link](#))
- **Traina et al. (2025):** "The total star formation rate function: UV and IR contributions up  $z \sim 6$  and comparison with state-of-the-art simulations" In *Astronomy & Astrophysics*, Volume 705, id.A255, 14 pp. ([ADS Link](#))

### 2025

- **Gentile et al. (2025a):** "Going deeper into the dark with COSMOS-Web: JWST unveils the total contribution of Radio-Selected NIRfaint galaxies to the cosmic Star Formation Rate Density" In *Astronomy & Astrophysics*, Volume 697, id.A46, 13 pp. ([ADS Link](#))
- **Akins et al. (2025):** "COSMOS-Web: The over-abundance and physical nature of "little red dots" — Implications for early galaxy and SMBH assembly" In *The Astrophysical Journal*, Volume 991, Issue 1, id.37, 21 pp. ([ADS Link](#))
- **Euclid Collaboration: Mellier et al. (2025):** "Euclid. I. Overview of the Euclid mission " In *Astronomy & Astrophysics*, Volume 697, id.A1, 94 pp ([ADS Link](#))
- **Delvecchio et al. (2025):** "AGN-heated dust revealed in "Little Red Dots" In *Astronomy & Astrophysics*, Volume 704, id.A313, 11 pp. ([ADS Link](#))
- **Faisst et al. (2025):** "COSMOS-Web: The Role of Galaxy Interactions and Disk Instabilities in Producing Starbursts at  $z < 4$ " In *The Astrophysical Journal*, Volume 980, Issue 2, id.204, 12 pp. ([ADS Link](#))
- **Guo et al. (2025):** "Ly-alpha emission reveals two satellite halos around massive groups at  $z \sim 3$ : the puzzling case of a quiescent central galaxy" In *Astronomy & Astrophysics*, Volume 704, id.A340, 19 pp. ([ADS Link](#))
- **McKinney et al. (2025):** "SCUBADive I: JWST+ALMA Analysis of 289 sub-millimeter galaxies in COSMOS-Web" In *The Astrophysical Journal*, Volume 979, Issue 2, id.229, 20 pp. ([ADS Link](#))
- **Shuntov et al. (2025a):** "COSMOS2025: The COSMOS-Web galaxy catalog of photometry, morphology, redshifts, and physical parameters from JWST, HST, and ground-based imaging" In *Astronomy & Astrophysics*, Volume 704, id.A339, 29 pp. ([ADS Link](#))
- **Toni et al. (2025):** "The COSMOS-Web deep galaxy group catalog up to  $z = 3.7$ " In *Astronomy & Astrophysics*, Volume 697, id.A197, 17pp. ([ADS Link](#))

### 2024

- **Gentile et al. (2024a):** "Illuminating the Dark Side of Cosmic Star Formation III: Building the largest homogeneous sample of Radio-Selected Dusty Star-Forming Galaxies in COSMOS with PhoEBO" In *The Astrophysical Journal*, Volume 962, id. 26, 16 pp ([ADS Link](#))
- **Gentile et al. (2024b):** "Dark progenitors and massive descendants: A first ALMA perspective on Radio-Selected NIRdark galaxies in the COSMOS field" In *Astronomy & Astrophysics*, Volume 687, id. A228, 18 pp ([ADS Link](#))
- **Gentile et al. (2024c):** "Not-so-little Red Dots: Two massive and dusty starbursts at  $z \sim 5 - 7$  pushing the limits of star formation discovered by JWST in the COSMOS-Web survey" In *The Astrophysical Journal Letters*, Volume 973, id. L2, 8 pp ([ADS Link](#))
- **Euclid Collaboration: Leuzzi et al. (2024):** "Euclid Preparation: TBD Characterization of Convolutional Neural Networks for the identification of Galaxy-Galaxy Strong Lensing events" In *Astronomy & Astrophysics*, Volume 681, id.A68, 23 pp ([ADS Link](#))
- **Langan et al. (2024):** "ALMA reveals a dust-obscured galaxy merger at Cosmic Noon" In *Astronomy & Astrophysics*, Volume 689, id.A283, 10 pp ([ADS Link](#))
- **Traina et al. (2024):** "A3COSMOS: the total infrared luminosity function and star formation rate density at  $z=0.5-6$ " In *Astronomy & Astrophysics*, Volume 681, id.A118, 16 pp ([ADS Link](#))



## Publications (continued)

---

### 2023

- **Gentile et al (2023):** "LEMON: LEns MOdelling with Neural networks – I. Automated modelling of strong gravitational lenses with Bayesian Neural Networks" In *Monthly Notices of the Royal Astronomical Society*, Volume 522, Issue 4, July 2023, Pages 5442–5455 [ADS Link](#)
- **Behiri et al (2023):** "Illuminating the Dark Side of Cosmic Star Formation II. A second date with RS-NIRdark galaxies in COSMOS" In *The Astrophysical Journal*, Volume 957, Issue 2, id.63, 14 pp. [ADS Link](#)
- **Busillo et al (2023):** "CASCO: Cosmological and Astrophysical parameters from Cosmological simulations and Observations - I. Constraining physical processes in local star-forming galaxies" In *Monthly Notices of the Royal Astronomical Society*, Volume 525, Issue 4, pp.6191–6213 [ADS Link](#)
- **Casey et al (2023):** "COSMOS-Web: An Overview of the JWST Cosmic Origins Survey" In *The Astrophysical Journal*, Volume 954, Issue 1, id.31, 32 pp. [ADS Link](#)
- **McKinney et al (2023):** "A Near-Infrared Faint, Far-Infrared-Luminous Dusty galaxy at  $z \sim 5$  in COSMOS-Web" In *The Astrophysical Journal*, Volume 956, Issue 2, id.72, 12 pp. [ADS Link](#)
- **Rojas et al (2023):** "The impact of human expert visual inspection on the discovery of strong gravitational lenses" In *Monthly Notices of the Royal Astronomical Society*, Volume 523, Issue 3, pp.4413–4430 [ADS Link](#)

### 2022

- **Gentile et al (2022):** Lenses In VoicE (LIVE): searching for strong gravitational lenses in the VOICE@VST survey using convolutional neural networks In *Monthly Notices of the Royal Astronomical Society*: Volume 510, Issue 1, pp.500–514. [ADS Link](#)

### 2021

- **Cantiello et al (2021):** The Fornax Deep Survey with VST. IX. Catalog of sources in the FDS area with an example study for globular clusters and background galaxies. In *Astronomy & Astrophysics*: Volume 639, id.A136, 23 pp. [ADS Link](#)